3-11 High central neuraxial block v.1

- Can occur with deliberate or accidental injection of local anaesthetic drugs into the subarachnoid space.
- Symptoms are – in sequence – hypotension and bradycardia – difficulty breathing – paralysis of the arms – impaired consciousness – apnoea and unconsciousness.
- Progression through this sequence can be slow or fast.

START

1. Reassure the patient – remember that they may be fully aware.
   - Plan to ensure hypnosis as soon as clinical situation permits.
2. Call for help and inform theatre team of the problem.
3. Treat airway and breathing:
   - Give 100% oxygen.
   - Chin lift / jaw thrust may suffice.
   - Consider supraglottic airway or tracheal intubation (Box A).
4. Treat circulatory insufficiency:
   - Give i.v. fluid by rapid infusion.
   - Elevate the legs. Do not use head-down tilt.
   - In obstetrics, relieve aorto-caval compression.
   - Bradycardia: give atropine or glycopyrrolate (Box B).
   - Hypotension: give metaraminol, phenylephrine or ephedrine (Box B).
   - CPR may be necessary to circulate drugs.
5. If the case is obstetric, consider expedited delivery of the baby to manage:
   - Risk to mother of unrelieved aorto-caval compression
   - Risk to fetus of impaired feto-placental oxygen delivery
6. Consider other causes that may mimic signs and symptoms, including (Box C):
   - Obstetric aorto-caval compression.
   - Local anaesthetic toxicity.
   - Embolism.
   - Vasovagal event.
   - Haemorrhage.
7. Plan ongoing care in a suitable location.

Box A: INDUCING ANAESTHESIA

- Consider reduced dose of hypnotic drug to avoid further hypotension. A full induction dose will not be necessary if the patient's consciousness is already impaired.
- Neuromuscular blockade may not be necessary for tracheal intubation if the patient is unconscious, paralysed and apnoeic.

Box B: DRUG DOSES

Bradycardia:

- Atropine: 0.6-1.2 mg
- Glycopyrrolate: 0.2-0.4 mg

Hypotension:

- Metaraminol: 1-2 mg boluses repeated
- Phenylephrine: 50-100 µg boluses repeated or by infusion
- Ephedrine: 6-12 mg boluses repeated up to max 30 mg (tachyphylaxis limits further usefulness)

Box C: CRITICAL CHANGES

- Cardiac arrest → **2-1**
- Hypotension → **2-4**
- Bradycardia → **2-6**
- Local anaesthetic toxicity → **3-10**

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